have been observed for nuclides very far from the stability line, namely *neutron emission* and *proton emission*. A few very rare decay modes like ^{12}C-emission have also been observed.

In the following, for convenience, we sometimes use an abbreviated form for decay reactions, as illustrated for the ^{238}U decay chain described in §1.3:

$$^{238}\mathrm{U}(\alpha)\,^{234}\mathrm{Th}(\beta^-)\,^{234}\mathrm{Pa}(\beta^-)\,^{234}\mathrm{U}(\alpha),\ \text{etc.}, \tag{5.1}$$

or, if half-lives and details are of importance:

$$^{238}\mathrm{U}(\alpha,\ 4.5\times10^{9}\ \mathrm{y})\,^{234}\mathrm{Th}(\beta^-,\ 24\ \mathrm{d})\,^{234\mathrm{m}}\mathrm{Pa}(\beta^-,\ \mathrm{IT},\ 1.17\ \mathrm{min})$$
$$^{234}\mathrm{Pa}(\beta^-,\ 6.7\ \mathrm{h})\,^{234}\mathrm{U}(\alpha,\ 2.5\times10^{5}\ \mathrm{y}),\ \text{etc.} \tag{5.2}$$

In the following chapter we discuss the energetics of the decay processes based on nuclear binding energy considerations and simple mechanics, then we consider the kinetics of the processes. In Ch. 6, where the internal properties of the nuclei are studied, the explanations of many of the phenomena discussed in this chapter are presented in terms of simple quantum mechanical rules.

5.2. CONSERVATION LAWS

In radioactive decay – as well as in other nuclear reactions – a number of *conservation laws* must be fulfilled. Such laws place stringent limitations on the events which may occur.

Consider the reaction

$$\mathrm{X}_1 + \mathrm{X}_2 \ \rightarrow\ \mathrm{X}_3 + \mathrm{X}_4 \tag{5.3}$$

where X represents any nuclear or elementary particle. In induced nuclear reactions X_1 may be the bombarding particle (e.g. a ^{4}He atom in a beam of α-particles) and X_2 the target atom (e.g. ^{14}N atoms), and X_3 and X_4 the products formed (e.g. ^{1}H and ^{17}O).

Sometimes only one product is formed, sometimes more than two. In radioactive decay several products are formed; reaction (5.3) is then better written $\mathrm{X}_1 \rightarrow \mathrm{X}_2 + \mathrm{X}_3$. For generality, however, we discuss the conservation laws for the case (5.3).

For the general reaction (5.3):

(a) The *total energy* of the system must be constant, i.e.

$$E_1 + E_2 = E_3 + E_4 \tag{5.4}$$

where E includes all energy forms: mass energy (§4.3), kinetic energy, electrostatic energy, etc.

(b) The *linear momentum*

$$p = mv \tag{5.5}$$

must be conserved in the system, and thus

$$p_1 + p_2 = p_3 + p_4 \tag{5.6}$$

The connection between kinetic energy E_{kin} and linear momentum is given by the relation

$$E_{\text{kin}} = p^2/(2m) \tag{5.7}$$

(c) The *total charge* (protons and electrons) of the system must be constant, i.e.

$$Z_1 + Z_2 = Z_3 + Z_4 \tag{5.8}$$

where the charge is in electron units.

(d) The *mass number* (number of nucleons) in the system must be constant, i.e.

$$A_1 + A_2 = A_3 + A_4 \tag{5.9}$$

(e) The *total angular momentum* p_{I} of the system must be conserved, i.e.

$$(p_{\text{I}})_1 + (p_{\text{I}})_2 = (p_{\text{I}})_3 + (p_{\text{I}})_4 \tag{5.10}$$

Since there exist two types of angular momentum, one caused by orbital movement of the individual nucleons and the other due to the intrinsic spin of the nucleons (internal angular momentum), a more practical formulation of (5.10) is

$$\Delta I = I_3 + I_4 - I_1 - I_2 \tag{5.11}$$

where I is the (total) *nuclear spin quantum number.* The quantum rule is

$$\Delta I = 0,\ \pm 1,\ \pm 2,\ \pm 3, \ldots \tag{5.12}$$

i.e. the change of nuclear spin in a reaction must have an integral value.

The three first laws are general in classical physics; the last two refer particularly to nuclear reactions. In Ch. 6 and 11 other conservation laws are discussed for nuclear reactions, but these are less important in radioactive decay.

5.3. ALPHA DECAY

5.3.1. Detection

Alpha particles cause extensive ionization in matter, as discussed in more detail in Ch 8. If the particles are allowed to pass into a gas, the electrons released by the ionization can be collected on a positive electrode to produce a pulse or current. *Ionization chambers* and *proportional counters* are instruments of this kind, which permit the individual counting of each α-particle emitted by a sample. Alpha particles interacting with matter may also cause molecular excitation, which can result in fluorescence. This fluorescence – or